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Practical -2

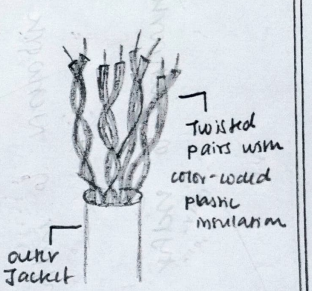
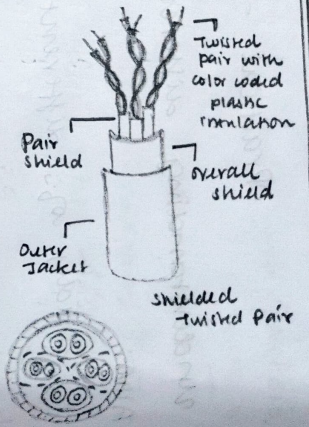
Understanding different types of network cable.

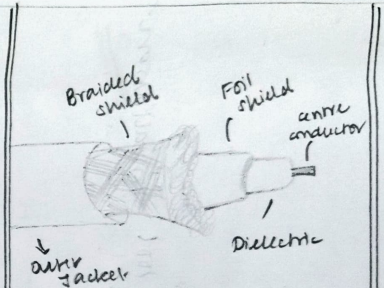
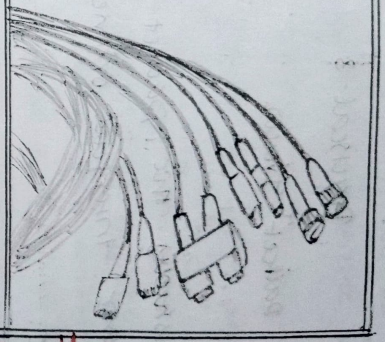
AIM:

Study of different types of Network cables.

Different types of cables used in networking are:

1) Unshielded twisted Pair (UTP) cable
2) Shielded twisted Pair (STP) cable
3) Coaxial cable
4) Fibre optic cable.

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / use	Image
UTP	Category 3	10bps	ADVANTAGES: • cheaper in cost • Easy to install - have small overall diameter DISADVANTAGES: • More prone to EMI Electromagnetic interference of noise	10 Base-T ethernet Fast Ethernet, Gigabit ethernet	
	Category 5	upto 100 Mbps			
	Category 5e	1Gbps			
STP	Category 6, 6a	10Gbps	ADVANTAGES: • Shielded • Faster than UTP • Less susceptible to noise and interference DISADVANTAGES: • Expensive • Greater installation effort	5gigabit Ethernet 10G Ethernet (55m) widely used in data centres	
SSTP	Category 7	10Gbps		Gigabit Ethernet 10G ethernet (100m)	

coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	ADVANTAGES • High bandwidth • Immune to interferen • Low loss bandwidth • versatile DISADVANTAGES • Limited distance • cost • Fibre is bulky	speed of signal is 500m Television network High speed internet connections	
Fibre optics cable	single mode Multi mode	100 Gbps	ADVANTAGES: • High speed • High bandwidth • High securing • long distance DISADVANTAGES: • Expensive • Requires skilled installers.	Maximum distance of fibre optics cable is around 100 m	

RESULT:

Thus we have studied the different types of network cables.